Vendor Performance Analysis

By:
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Introduction

This report presents a **Vendor Performance Analysis**, designed to transform raw sales, inventory, and vendor data into actionable business insights. The purpose of this project is to evaluate vendor and brand performance, identify top-performing and underperforming vendors, and uncover opportunities to improve profitability and inventory management.

To perform this analysis, I used Python for data cleaning and exploratory analysis, SQL for querying and aggregating datasets, and Power BI to develop interactive dashboards. The analysis highlights key trends and metrics to support strategic business decisions.

Business Problem

- Retail and wholesale companies often lose profits due to poor pricing, slow-moving inventory, and heavy dependency on a few vendors.
- This project analyzes vendor, sales, and inventory data to identify performance gaps and improve overall profitability.

Objectives

- Identify low-performing brands and suggest pricing or promotion changes.
- Find top vendors contributing the maximum sales and profit.
- Analyze the impact of bulk purchasing on unit cost.
- Improve inventory turnover to reduce holding costs.
- Compare profit margins between high-performing and low-performing vendors.

Dataset Overview

Tables and Records:

- vendor_invoice – 5,544 records
- sale – 12,825,363 records
- purchases – 2,372,474 records
- purchase_prices – 12,261 records
- begin_inventory – 206,529 records
- end_inventory – 224,489 records

Key Columns:

- Sales data: e.g., SalesPrice, SalesDollars (names may vary slightly in your dataset)
- Purchase data: e.g., PurchasePrice, Quantity
- Vendor/Brand identifiers: e.g., VendorName, Brand
- Inventory metrics: e.g., InventoryQty, UnsoldInventoryValue

Tools & Technologies

I. **Python** – Used Pandas and NumPy for data cleaning and manipulation, Matplotlib and Seaborn for exploratory data analysis.

II. **SQL** – Performed joins, aggregations, and queries to combine and summarize datasets.

III. **Power BI** – Developed interactive dashboards, visualizations, and applied DAX formulas for KPIs.

IV. **SQLite / CSV** – Data storage and extraction from multiple transactional tables.

Data Cleaning & Preparation

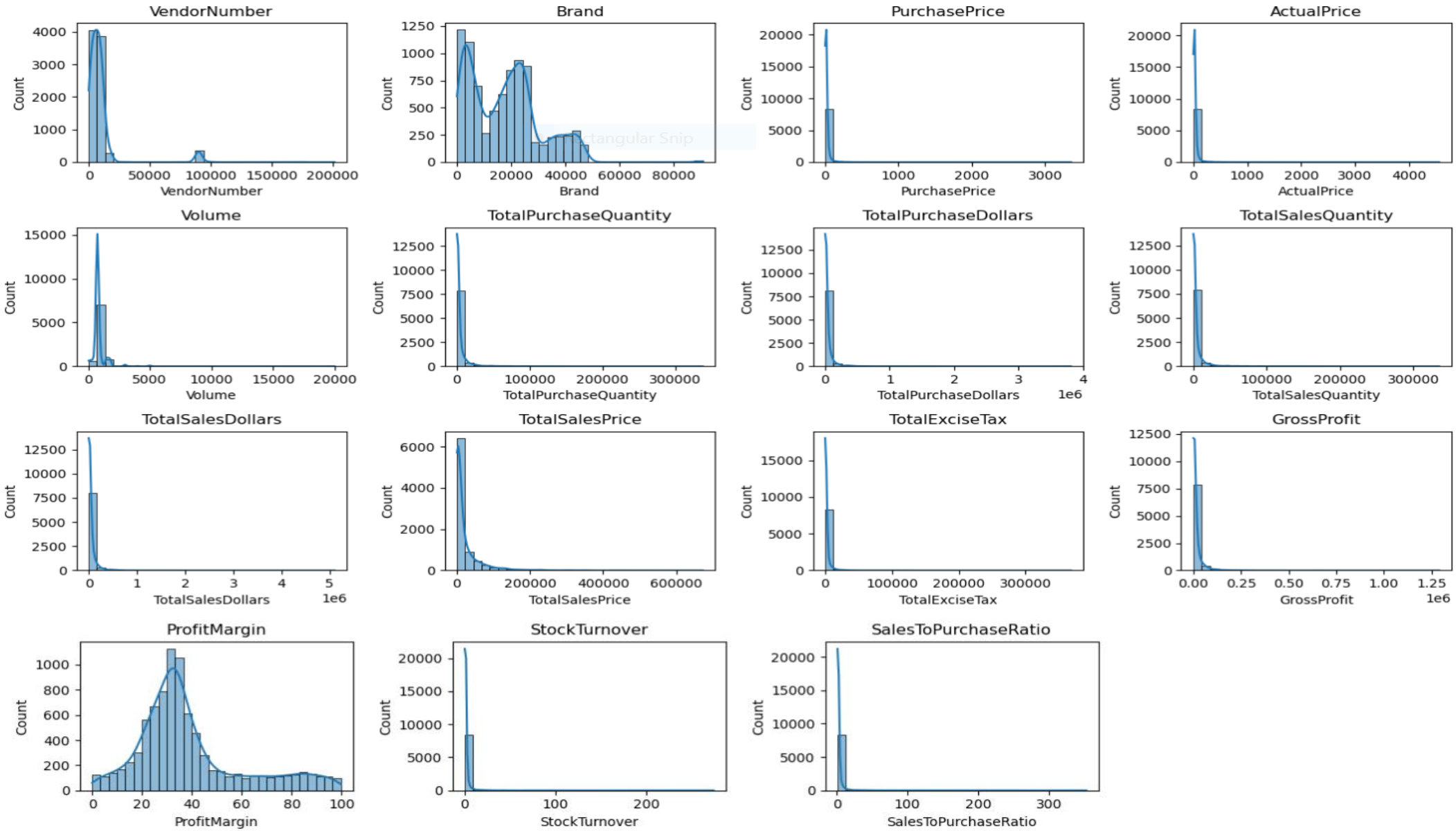
- Removed null and erroneous values to ensure data quality.
- Standardized column names across all datasets for consistency.
- Merged sales, purchase, and inventory tables into a unified dataset for analysis.
- Created a vendor_sales_summary table to aggregate metrics such as total sales, gross profit, profit margin, and purchase contribution per vendor.

Summary Statistics

	VendorNumber	VendorName	Brand	Description	PurchasePrice	ActualPrice	Volume	TotalPurchaseQuantity	TotalPurchaseDollars	TotalSalesQuantity	TotalSalesDollars	TotalSalesPrice	TotalExciseTax	GrossProfit	ProfitMargin	StockTurnover	SalesToPurchaseRatio
0	1128	BROWN-FORMAN CORP	1233	Jack Daniels No 7 Black	26.27	36.99	1750.0	145080	3811251.60	142049.0	5101919.51	672819.31	260999.20	1290667.91	25.297693	0.979108	1.338647
1	4425	MARTIGNETTI COMPANIES	3405	Tito's Handmade Vodka	23.19	28.99	1750.0	164038	3804041.22	160247.0	4819073.49	561512.37	294438.66	1015032.27	21.062810	0.976890	1.266830
2	17035	PERNOD RICARD USA	8068	Absolut 80 Proof	18.24	24.99	1750.0	187407	3418303.68	187140.0	4538120.60	461140.15	343854.07	1119816.92	24.675786	0.998575	1.327594
3	3960	DIAGEO NORTH AMERICA INC	4261	Capt Morgan Spiced Rum	16.17	22.99	1750.0	201682	3261197.94	200412.0	4475972.88	420050.01	368242.80	1214774.94	27.139908	0.993703	1.372493
4	3960	DIAGEO NORTH AMERICA INC	3545	Ketel One Vodka	21.89	29.99	1750.0	138109	3023206.01	135838.0	4223107.62	545778.28	249587.83	1199901.61	28.412764	0.983556	1.396897
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
8559	9815	WINE GROUP INC	8527	Concannon Glen Ellen Wh Zin	1.32	4.99	750.0	2	2.64	5.0	15.95	10.96	0.55	13.31	83.448276	2.500000	6.041667
8560	8004	SAZERAC CO INC	5683	Dr McGillicuddy's Apple Pie	0.39	0.49	50.0	6	2.34	134.0	65.66	1.47	7.04	63.32	96.436186	22.333333	28.059829
8561	3924	HEAVEN HILL DISTILLERIES	9123	Deep Eddy Vodka	0.74	0.99	50.0	2	1.48	2.0	1.98	0.99	0.10	0.50	25.252525	1.000000	1.337838
8562	3960	DIAGEO NORTH AMERICA INC	6127	The Club Strawbry Margarita	1.47	1.99	200.0	1	1.47	72.0	143.28	77.61	15.12	141.81	98.974037	72.000000	97.469388
8563	7245	PROXIMO SPIRITS INC.	3065	Three Olives Grape Vodka	0.71	0.99	50.0	1	0.71	86.0	85.14	33.66	4.46	84.43	99.166079	86.000000	119.915493

8564 rows × 17 columns

Histogram Plot



Summary Statistics & Data Observations

Negative & Zero Values:

- Gross Profit: Minimum of 52,002.78, indicating potential high costs or heavy discounts. This could be due to selling products at lower prices than their purchase costs.
- Profit Margin: Has a minimum of ∞, which suggests instances where revenue is zero or even lower than the total cost, leading to extreme negative profit margins.
- Total Sales Quantity & Sales Dollars: Some products show zero sales, indicating they were purchased but never sold. These may be slow-moving or obsolete stock, leading to inventory inefficiencies.

Outliers Detected by High Standard Deviations:

- Purchase & Actual Prices: The maximum values (5,681.81 & 7,499.99) are significantly higher than the mean (24.39 & 35.64), indicating premium product offerings.

Freight Cost:

Extreme variation from 0.09 to 257,032.07 suggests logistics inefficiencies, bulk shipments, or erratic shipping costs across different products.

Stock Turnover:

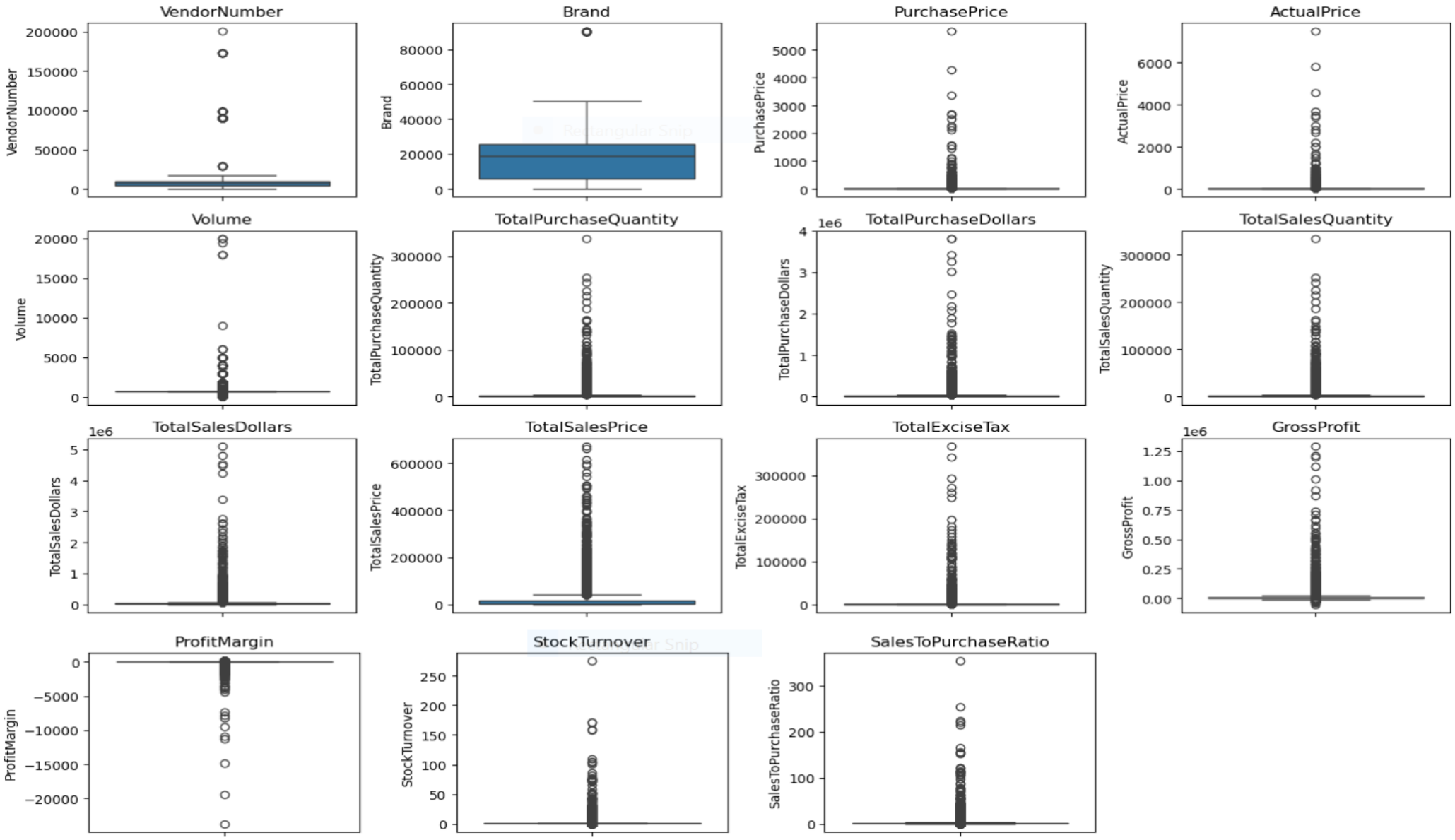
Ranges from 0 to 274.5, suggesting some products sell rapidly while others remain unsold for long periods. A value greater than 1 indicates that sales for a product exceed the purchased quantity due to older stock fulfilling orders.

Data Filtering

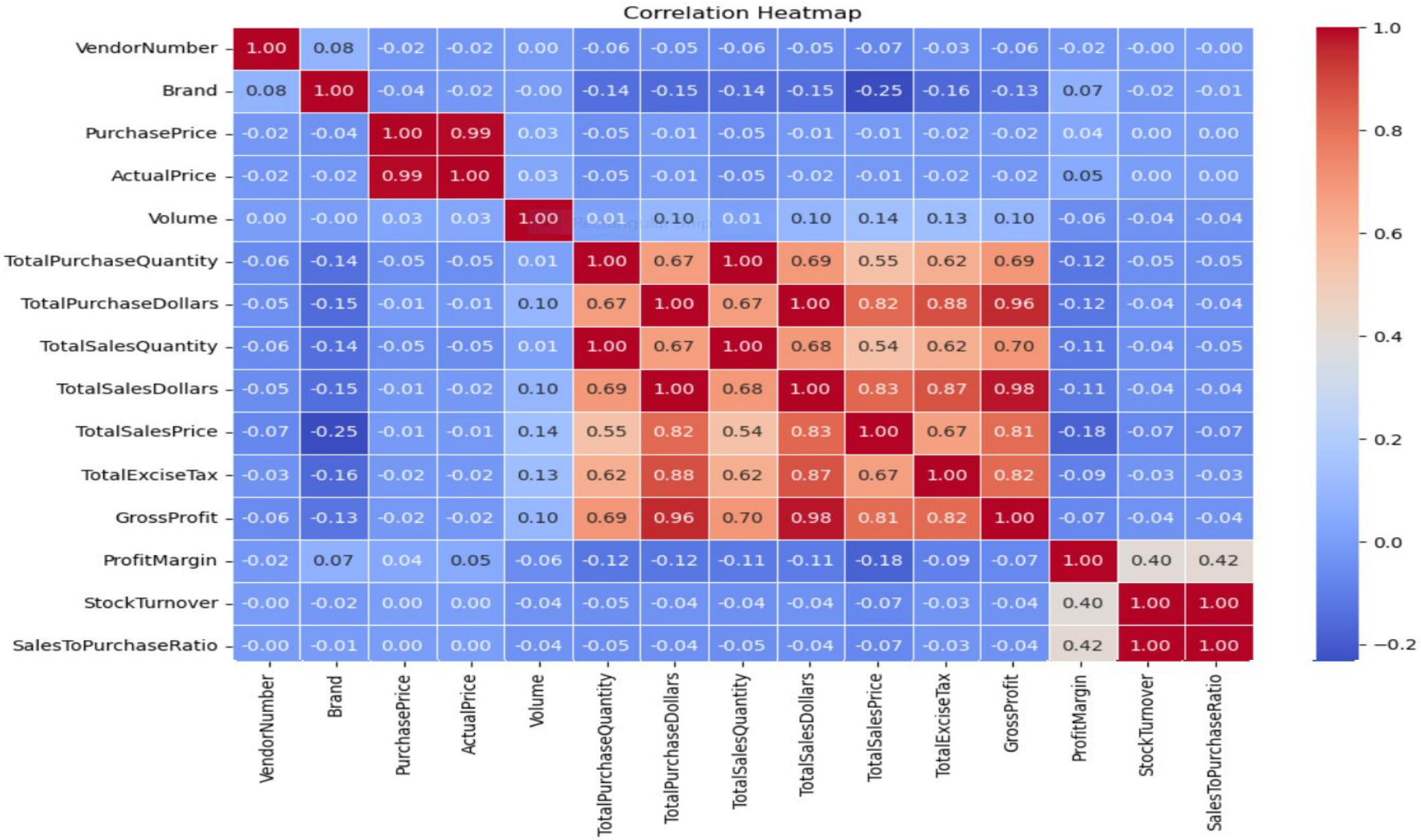
To enhance the reliability of the insights, we removed inconsistent data points where:

- Gross Profit ≤ 0 (to exclude transactions leading to losses).
- Profit Margin ≤ 0 (to ensure analysis focuses on profitable transactions).
- Total Sales Quantity = 0 (to eliminate inventory that was never sold).

Finding Outliers



Finding Outliers



Correlation Analysis & Insights

Purchase Price vs. Total Sales Dollars & Gross Profit: Weak negative correlation (-0.012 and -0.016), indicating that price variations do not significantly impact sales revenue or profit.

Total Purchase Quantity vs. Total Sales Quantity: Strong positive correlation (0.999), confirming highly efficient inventory turnover.

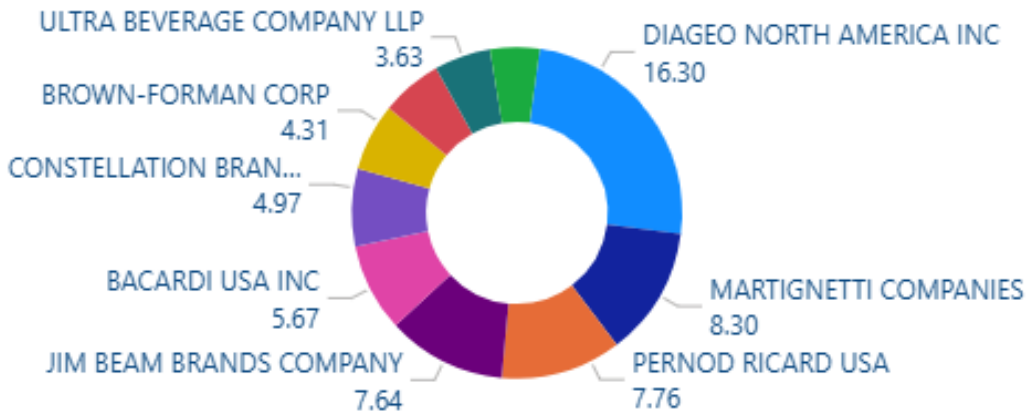
Profit Margin vs. Total Sales Price: Negative correlation (-0.179), suggesting that increasing sales prices may lead to reduced margins, possibly due to competitive pricing pressures.

Stock Turnover vs. Gross Profit & Profit Margin: Weak negative correlation (-0.038 and -0.055), indicating that faster stock turnover does not necessarily result in higher profitability.

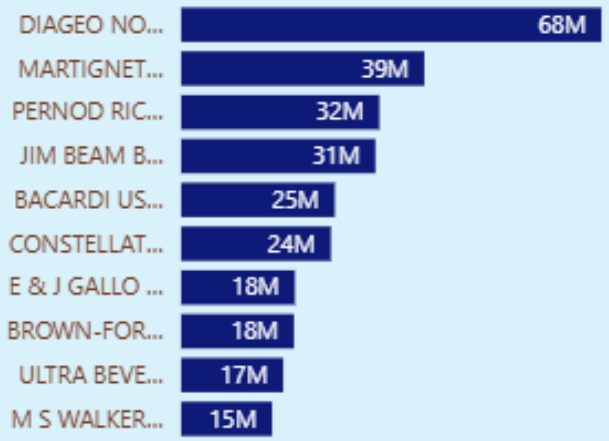
Vendor Performance Dashboard

441.41M	307.34M	134.07M	38.72	2.71M
Total Sales ($)	Total Purchase ($)	Gross Profit ($)	Profit Margin(%)	Unsold Capital ($)

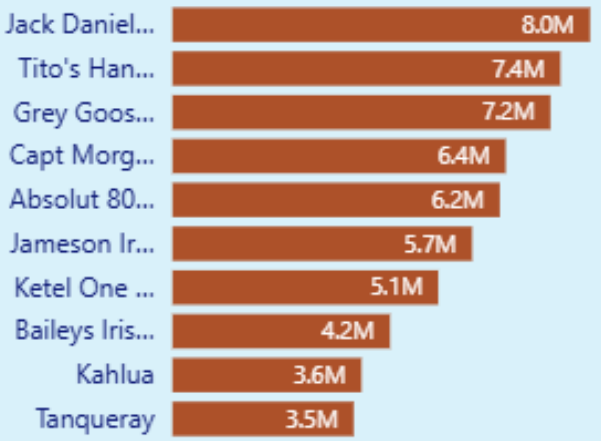
Purchase Contribution%



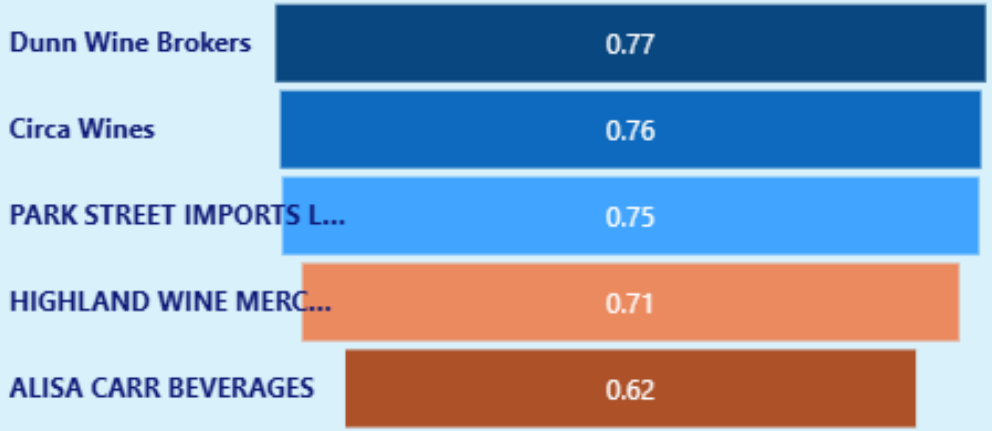
Top Vendors By Sales



Top Brands By Sales



Low Performing Vendors



Low Performing Brands

Insights:

- Weak relationship between price and profit
- Strong relationship between purchase and sales quantity (0.999)
- Higher prices slightly reduce margins
- Faster turnover does not guarantee higher profit

Research Questions and Key Findings

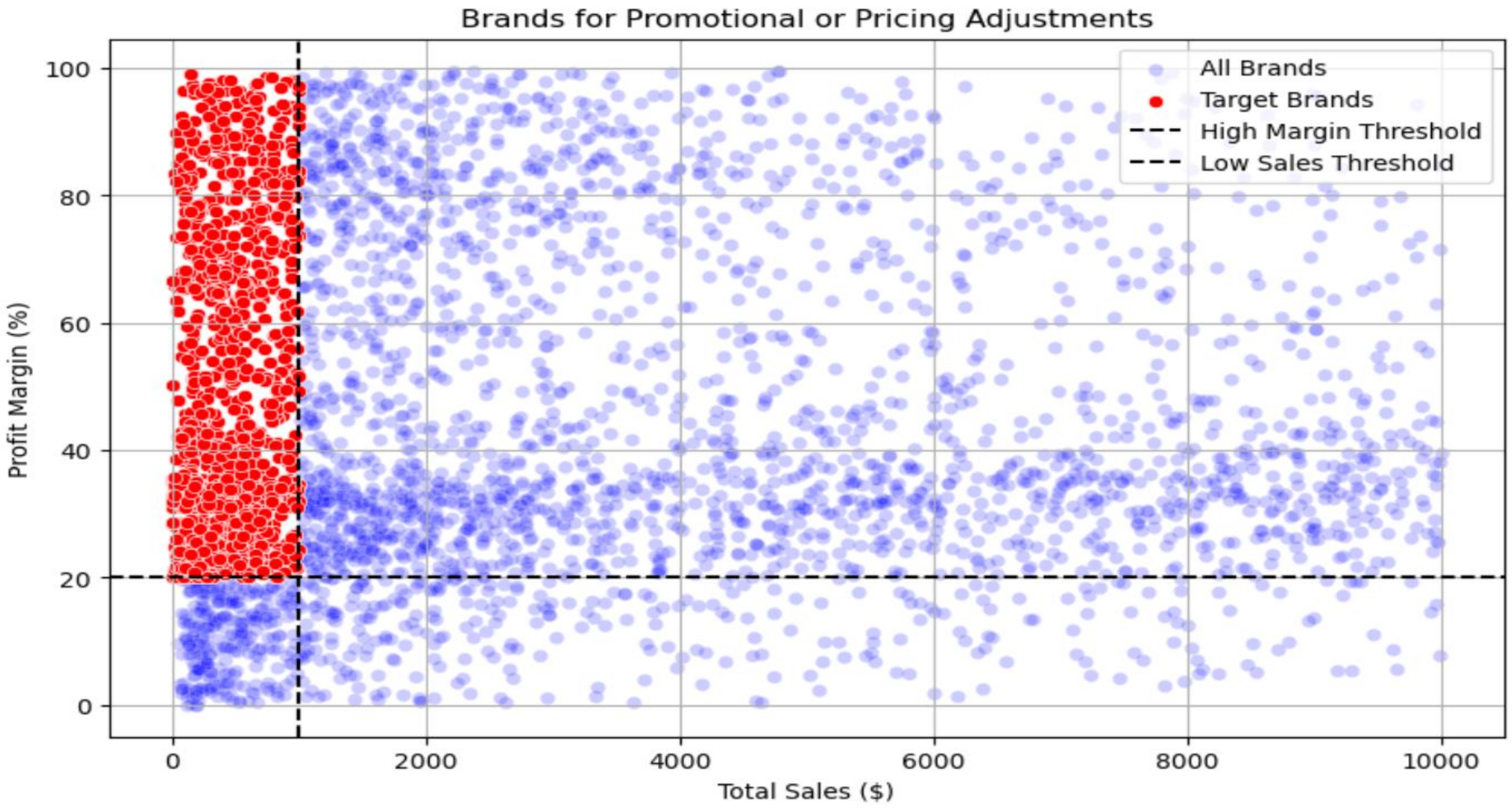
1. Brands For Promotional And Pricing Adjustments

Brands with Low Sales but High Profit Margins:

	Description	TotalSalesDollars	ProfitMargin
6942	Terruzzi & Puthod Vernaccia	4.99	32.464930
2897	Firefly Ridge Chard	5.79	31.088083
148	Albero Sparkling Wh Organic	5.79	28.497409
6488	Smirnoff Light Strawberry	5.94	50.168350
178	Allen's Blue Curacao Liqueur	5.99	20.033389
...	...	...	...
3476	Hartford Court Chard RRV	998.61	49.259471
6687	Steph Avrn Chn Vielles Vig	999.50	97.278639
4685	Marques de Grinon 10 Cab Svg	999.75	83.783946
3125	Galerie Latro Cab S Knights	999.80	93.332667
1787	Ch de Beaucastel Pape Blanc	999.92	82.876630

1368 rows × 3 columns

Brands For Promotional and Pricing Adjustments

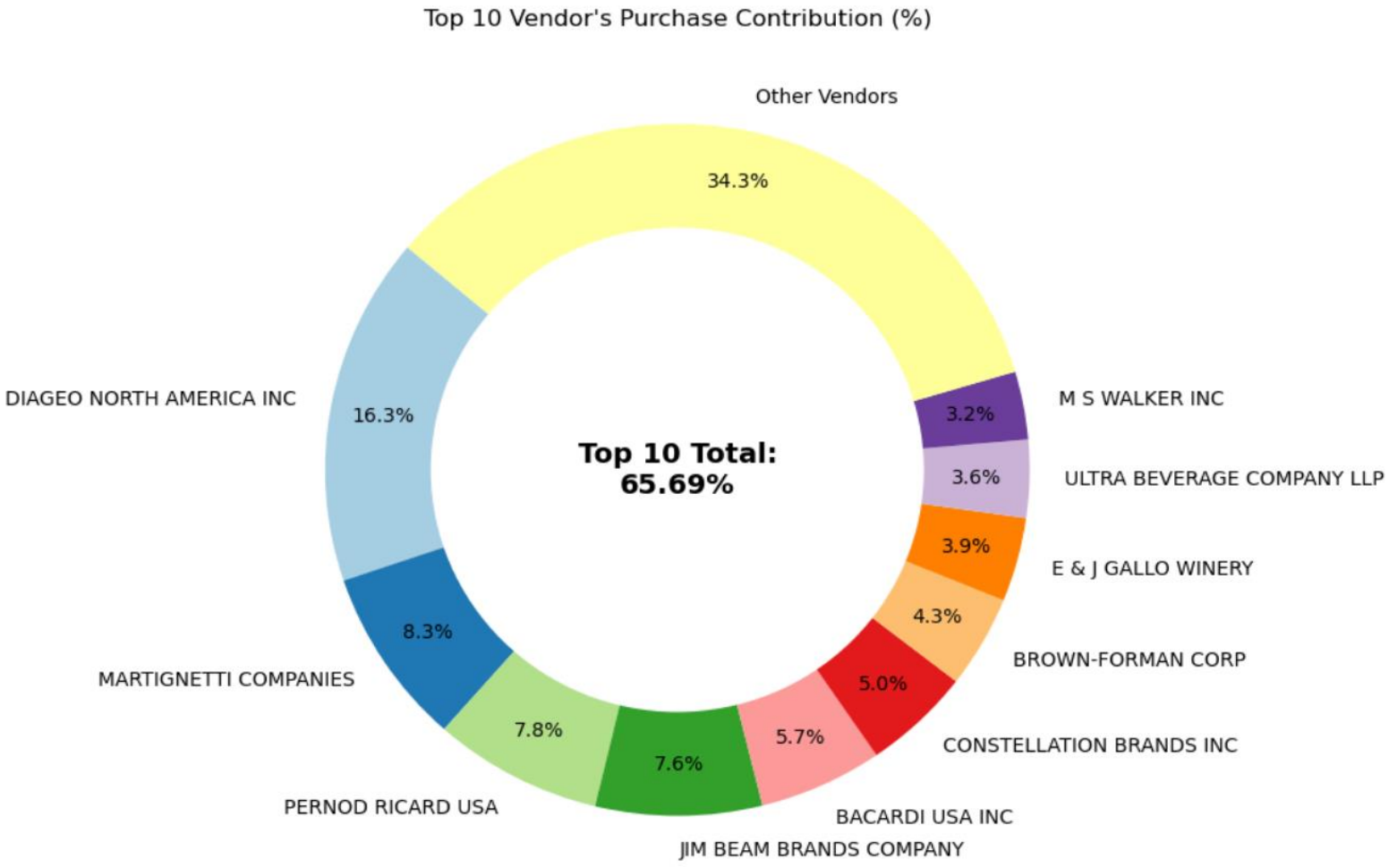


Insights:

198 brands exhibit lower sales but higher profit margin, which could be benefit for target marketing, Promotion or price optimizations to increase volume without compromising profitability

2. Top Vendors By Sales & Purchase Contribution:

The top 10 vendors contribute 65.69% of total purchases, while the remaining vendors contribute only 34.31%. This over-reliance on a few vendors may introduce risks such as supply chain disruptions, indicating a need for diversification.



3. Impact of Bulk Purchasing on Cost Savings

Vendors buying in large quantities receive a 72% lower unit cost ($10.78 per unit vs. higher unit costs in smaller orders).

Bulk pricing strategies encourage larger orders, increasing total sales while maintaining profitability.

OrderSize	UnitPurchasePrice
Small	39.068186
Medium	15.486414
Large	10.777625

4. Identifying Vendors with Low Inventory Turnover

Total Unsold Inventory Capital: $2.71M

Slow-moving inventory increases storage costs, reduces cash flow efficiency, and affects overall profitability.

Identifying vendors with low inventory turnover enables better stock management, minimizing financial strain.

VendorName	StockTurnover
ALISA CARR BEVERAGES	0.615385
HIGHLAND WINE MERCHANTS LLC	0.708333
PARK STREET IMPORTS LLC	0.751306
Circa Wines	0.755676
Dunn Wine Brokers	0.766022
CENTEUR IMPORTS LLC	0.773953
SMOKY QUARTZ DISTILLERY LLC	0.783835
TAMWORTH DISTILLING	0.797078
THE IMPORTED GRAPE LLC	0.807569
WALPOLE MTN VIEW WINERY	0.820548

	VendorName	UnsoldInventoryValue
25	DIAGEO NORTH AMERICA INC	$722,209.05
46	JIM BEAM BRANDS COMPANY	$554,665.63
68	PERNOD RICARD USA	$470,625.61
116	WILLIAM GRANT & SONS INC	$401,960.83
30	E & J GALLO WINERY	$228,282.61
79	SAZERAC CO INC	$198,436.41
11	BROWN-FORMAN CORP	$177,733.74
20	CONSTELLATION BRANDS INC	$133,617.62
61	MOET HENNESSY USA INC	$126,477.70
77	REMY COINTREAU USA INC	$118,598.15

5. Profit Margin Comparison: High vs. Low-Performing Vendors

- Top Vendors' Profit Margin (95% CI): (30.74%, 31.61%), Mean: 31.17%
- Low Vendors' Profit Margin (95% CI): (40.48%, 42.62%), Mean: 41.55%
- Low-performing vendors maintain higher margins but struggle with sales volumes, indicating potential pricing inefficiencies or market reach issues.

Profit Margin Comparison

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Top Vendors 95% CI: (30.74, 31.61), Mean: 31.18
Low Vendors 95% CI: (40.50, 42.64), Mean: 41.57
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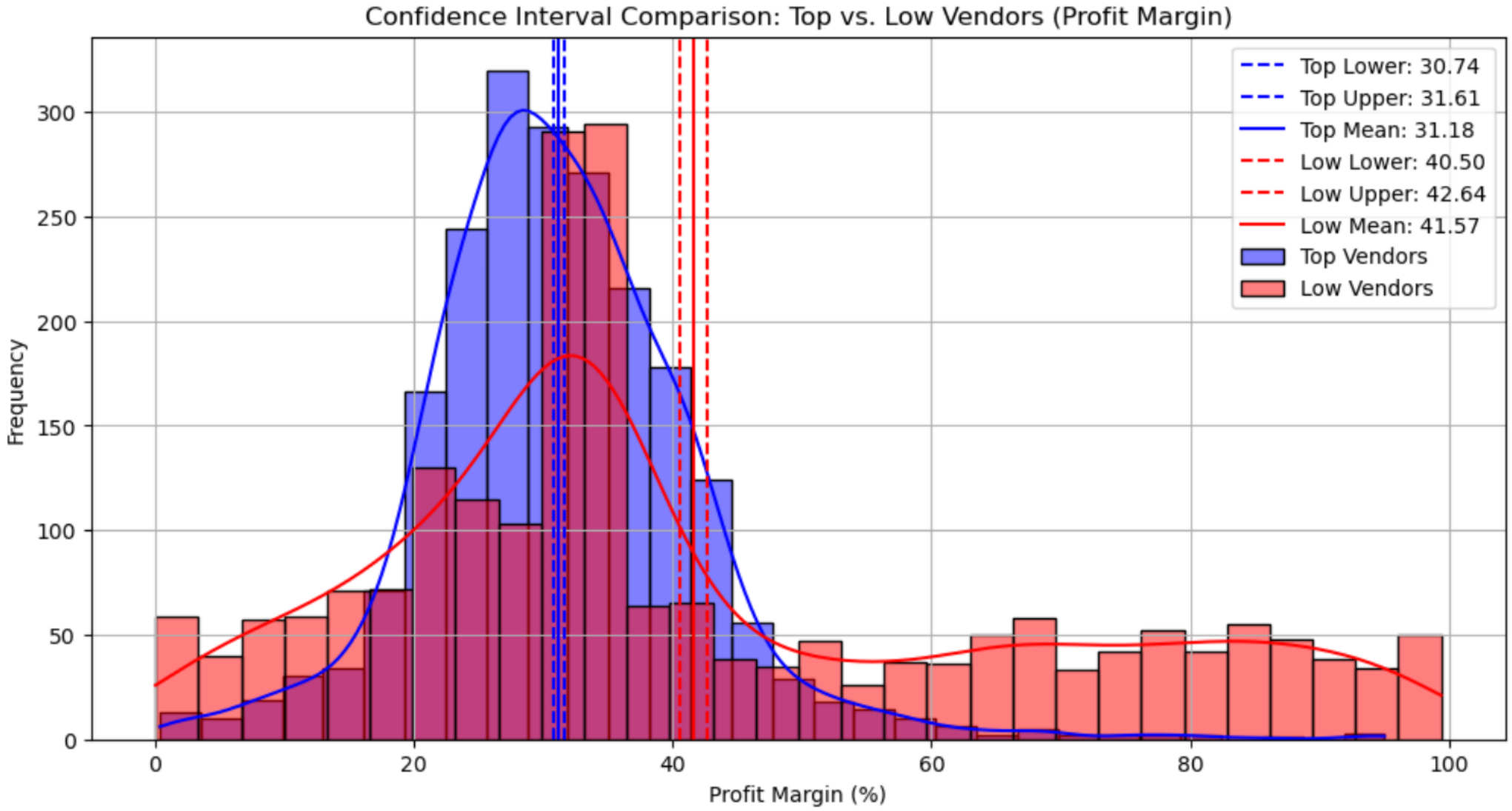

Statistical Validation of Profit Margin Differences

Hypothesis Testing:

Ho (Null Hypothesis): No significant difference in profit margins between top and low-performing vendors.

H. (Alternative Hypothesis): A significant difference exists in profit margins between the two vendor groups.

Result: The null hypothesis is rejected, confirming that the two groups operate under distinctly different profitability models.

Implication: High-margin vendors may benefit from better pricing strategies, while top-selling vendors could focus on cost efficiency.

FINAL RECOMMENDATION

- ❑Adjust pricing for low-volume, high-margin brands
- ❑Reduce dependency on few vendors
- ❑Use bulk purchasing strategically
- ❑Clear slow-moving inventory Improve marketing for low-performing vendors